FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

3 Week: 27 February 2017 / 27 Februarie 2017

All the inidcated numbers refer to the "Problems" section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die "Problems" afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P4.8 (a) $(10.0\hat{\imath} + 0.241\hat{\jmath})$ mm (b) $(1.84 \times 10^7$ m/s$)\hat{\imath} + (8.78 \times 10^5$ m/s$)\hat{\jmath}$
(c) 1.85×10^7 m/s (d) $2.73°$
P4.9 (a) $a_x = 0.800$ m/s^2; $a_y = -0.300$ m/s^2 (b) $\theta = 339°$ from $+x$-axis
(c) $x_f = 360$ m; $y_f = -72.7$ m; $\theta = -15.2°$
P4.15 $\theta_i = 53.1°$
P4.21

$$h = d\tan\theta_i - \frac{gd^2}{2v_i^2\cos^2\theta_i}$$

P4.28 (a)

$$\frac{v_i\sin\theta}{g}$$

(b)

$$h + \frac{(v_i\sin\theta)^2}{2g}$$

P4.29 (a) $x_i = 0.00$ m; $y_i = 0.00$ m (b) $v_{xi} = 18.0$ m/s; $v_{yi} = 0$ m
(c) Free fall motion, with constant downward acceleration, $g = 9.80$ m/s^2
(d) Constant velocity motion in the horizontal direction (e) $v_{xf} = v_{xi}$; $v_{yf} = -gt$ (f) $x_f = v_{xi}t$; $y_f = -\frac{1}{2}gt^2$ (g) 3.19 s (h) $v_f = 36.1$ m/s; $\theta_f = -60.1°$
P4.58 (a) $\vec{v} = (5\hat{\imath} + 4t^{3/2}\hat{\jmath})$ m/s (b) $\vec{r} = (5t\hat{\imath} + 1.6t^{5/2}\hat{\jmath})$ m
P4.61 (a) $v = 26.9$ m/s (b) $d = 67.3$ m (c) $\vec{a} = (2.00\hat{\imath} - 5.00\hat{\jmath})$ m/s^2
P4.65 (a) 1.52×10^3 m (b) 36.1 s (c) 4.05×10^3 m
P4.73 (a) $x(v_i) = (v_i\sqrt{0.1643 + 0.002299v_i^2} + 0.04794v_i^2)$ m (b) 0.0410 m
(c) 961 m (d) 0.405 v_i (e) 0.0959 v_i^2 (f) Explanation
P4.74 (a) $\theta = 26.6°$ (b) 0.949